Rajiv Choudhury

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Summary

MS Computer Science student at **UC Davis** and B.Tech graduate from **IIIT Guwahati** (CPI: 9.13/10), with a strong foundation in **C++, Python, Data Structures, and Algorithms**. Experienced in **peer instruction, technical mentoring, and software engineering**, including leading a six-member undergraduate C++ study group and mentoring interns at Samsung R&D. Passionate about helping students develop programming proficiency and problem-solving skills.

Education

Graduate Student — MS (Computer Science) at **UC Davis** (Sept 2026 - Jun 2028 (*ten.*))
***Relevant Coursework**: Programming Languages, Large-scale Scientific Computation*

Class of 2024 — B.Tech (Computer Science Engineering) at **IIIT Guwahati** (CPI: 9.13/10)
***Relevant Coursework**: Data Structures & Algorithms, Software Engineering, Cloud Computing, Operating Systems*

Teaching & Mentorship Experience

TA for ECS 130 — UC Davis Fall 2026

– Grading quizzes, homework assignments, and holding office hours for doubts for the course ECS 130.

C++ Programming Study Group Lead — IIIT Guwahati Undergraduate

– Led a peer study group of **6 undergraduate students**, facilitating collaborative learning sessions focused on **C++ programming**.

– Explained programming concepts, guided peers through coding exercises, and provided individualized support to strengthen their understanding of C++.

Work Experience

Software Engineer — Samsung R&D Institute India - Delhi Jul 2024 – Aug 2026

– Conducted **technical tutorial sessions** for team members, presenting new concepts related to product technologies, including **Agentic AI** and **Retrieval-Augmented Generation (RAG)**.

– **Mentored five interns** during their onboarding, providing **technical guidance** on product architecture and functionality to streamline their learning process.

– Developed and maintained **Tizen OS** features using **C++** and **C#**, including zero-contact USB presets for **E-paper signage**, certificate-expiration warnings for **10,000+ enterprise displays**, and offline **OpenAI Whisper** speech-to-text integration on Android Displays; contributed to code reviews and technical documentation, earning the team the **Best Feature Award**.

Research Intern — Centre for Networked Intelligence, IISc Bengaluru Jan 2024 – Jun 2024

– Conducted research on **NVIDIA GPU architecture** and parallel computing, distilling complex HPC concepts into accessible technical summaries.

– Proposed and documented a scheduling method achieving up to **80% improvement** in energy efficiency over standard Linux CPU governors. (Published in **ACM eEnergy** conference 2024 Singapore)

– Collaborated with **IBM Research** on GPU scheduling and matrix multiplication optimization by implementing and benchmarking **CUDA C++** machine learning models against **PyTorch**.

Research Intern — IIIT Guwahati May 2022 - Jul 2022

– Worked on **Brain-Computer Interface (BCI)** systems, conducting comprehensive literature reviews on EEG signal classification.

– Implemented **MLP classifiers and Neural Networks**, communicating technical research findings and methodologies effectively. (Published this work in **IEEE DCOSS-IoT** Conf. Abu Dhabi 2024)

Skills

Languages	Python, C++, CUDA C++, Java, C#, C
Web & Publishing	Google Colab, Jupyter Book, Web Accessibility Principles
Tools & Platforms	Git, GitHub, Unix/Linux Command Line, Docker, AWS, Android, Tizen
Data & AI Tools	Google Gemini, OpenAI Whisper, NumPy, Pandas, Scikit-learn
Frameworks & DBs	PyTorch, TensorFlow, Jetpack Compose, MySQL, MongoDB